

Advanced Topics in Databases: Final Report

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0.1 Introduction

Everyday, we are bombarded with a sheer amount of data that has to be stored in databases and efficiently managed to extract meaningful insights.

0.2 Development

So, for this project, we used the 2021 Portuguese local elections as the basis.

0.2.1 Functions, Views, and Triggers

- **Functions:** We created several functions to perform various operations on the election data, such as:

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- Calculating vote percentage for a candidacy.
- Calculating total votes for a party.
- Getting top N parties by vote percentage.
- Using the D'Hondt method to allocate seats.

- **Views:**

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- **Triggers:**

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0.2.2 Analytical Queries

After implementing the functions, views, and triggers, we tested them using SQL queries to ensure they were working correctly and efficiently. We also analyzed the results to gain insights into the election data, such as identifying which parties had the most votes and how the seats were allocated based on the D'Hondt method.

0.2.3 Frontend Overview

0.2.4 Known Limitations

0.3 Conclusion

.1 Appendix